

Prague University of Economics and Business

Faculty of Informatics and Statistics



SPECIAL FEATURES AND DATA STREAMS OF OWNER CONSENT FOR FIBRE ROLL-OUT

BACHELOR THESIS

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Declaration

I declare that this thesis entitled Special features and data streams of owner consent for fibre roll-out is a result of my own research except as cited in the references. All used literature and materials are mentioned in the attached list of bibliography.

In Prague 30th of April 2024

.....
Marek Lukl

Acknowledgement

First, I'd like to thank the supervisor of this bachelor thesis, doc. Ing. Jan Pour, CSc., who offered a solution to every complication that occurred. He guided me through by providing quick feedback on my progress and answers to all of my questions. Second, I'm grateful to my colleagues and especially our product owner from the telecom company, who always went the extra mile to support me with business and technical explanations of our data architecture.

Last but not least, I want to show gratitude to my family who have supported me throughout my studies.

Abstract

This thesis aims to investigate the approvals for fiber optic connections by a telecom company, specifically focusing on connections to basements versus connections to living units. The primary objective is to quantify and compare the number of approvals granted for these two types of connections. This new data will be then used for calculations of requested KPIs. In addition to this analysis, the study will provide a comprehensive overview of the telecommunications industry, with a specific emphasis on fiber optic technology in Germany, its evolution, and comparison to previous technologies. The research will also explore the data journey through various layers within a data architecture, including acquisition, staging, the core layer, and ultimately the export layer. Furthermore, the thesis will involve the development and implementation of a daily updated dashboard to visualize and present the findings effectively. This project will contribute to advancing knowledge in the analysis of telecommunications data and demonstrate practical applications of data processing and visualization techniques in a real-world industry context.

Keywords

data layering, fiber optic, data architecture, owner consent for fiber roll-out

Abstrakt

Tato bakalářská práce si klade za cíl prozkoumat schvalování připojení optického internetu telekomunikační společností, s konkrétním zaměřením na připojení do sklepů oproti připojení k bytovým jednotkám. Primárním cílem je kvantifikovat a porovnat počet udělených schválení pro tyto dva typy připojení, nová data pak mohou být použita pro výpočty požadovaných KPI. Kromě této analýzy studie poskytne přehled telekomunikačního odvětví, s konkrétním důrazem na technologii optického internetu v Německu, jeho vývoj a srovnání s předchozími technologiemi. Výzkum rovněž prozkoumá cestu dat skrze různé vrstvy datové architektury, včetně akvizice, přípravné vrstvy, core vrstvy, a nakonec exportní vrstvy. Kromě toho se práce bude zabývat vývojem a implementací denně aktualizovaného dashboardu pro vizualizaci a efektivní prezentaci zjištěných poznatků. Tento projekt přispěje k rozvoji poznatků o analýze telekomunikačních dat a demonstraci praktické aplikace postupů zpracování a vizualizace dat v kontextu skutečného podnikatelského prostředí.

Klíčová slova

datové vrstvy, optický internet, datová architektura, souhlas vlastníka pro zavedení optiky

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Introduction

Topic Definition

The topic of the thesis is an analysis of property owner decisions regarding fiber deployment. In the rapidly evolving world of telecommunications, the deployment of fiber optic connections to buildings has become one of the biggest topics for reaching competitive levels of digitalization in many countries. This thesis focuses on Germany, which in recent years substantially changed its approach to fiber roll-out and now is in the middle of an intensive transition to reach its ambitious goals for nationwide fiber connection. Such an operation requires cooperation from landlords of the affected buildings and mainly their approval either for installation of the technology just to the basement or to every living unit. It's this distinction between different types of property owner approvals that the focus is on.

As companies started with this new fiber project, they had to develop many procedures and systems to manage upcoming challenges. These systems generate substantial amounts of data, which plays a critical role in providing a comprehensive overview of the project's progress and supporting data-driven decision-making with high-quality information. This thesis investigates the task of processing data regarding fiber roll-out. It examines the characteristics and purpose of every data layer, first in theory and then through the lens of a big data team in one of the biggest telecommunications companies in Germany.

Reasons for Choosing the Topic

As a data engineer in an agile team, most of your work consists of unrelated tasks, fixing concrete steps in the data process, and expanding the application. It's a little bit like a construction worker working on a huge complex house. You focus on certain parts you work on but don't necessarily understand the entire process of getting the data/materials and all the steps involved to get to the end results. This thesis helped me to not only understand all the steps required but also the reasoning behind each one of them.

A further motivational factor is that especially the practical part and outcomes of this thesis are useful to a real-world company, which means I could allocate more time to writing it because it was partially classified as work. The last reason has something to do with the language of the thesis being English, even though it's not my native language. This decision

is about improving my ability to write scientific papers, which would help me in case I ever get to fulfill my dream of pursuing a master's degree in Scandinavia.

Goals of the Thesis

The goal of the thesis is to discover how many approvals does the telecom company have for fiber connection just to the basement and how many to the living units. This data can be then used for the calculation of new KPIs which should be incorporated into the form of a dashboard.

Partial goals are to analyze the telecommunications industry, especially in Germany, with a focus on fiber optic technology compared to its predecessor. Then an examination of the whole data journey through different layers firstly from a theoretical perspective and afterwards building the processing required.

Approach to the Solution

To successfully reach the goal of the thesis I used the expertise provided by my experience in the field of data engineering and telecommunications. Furthermore, some of the theoretical knowledge acquired during my studies at the Prague University of Economics and Business ended up being useful when analyzing the data journey.

Another important factor was frequent meetings with more experienced data engineers and architects who had been part of the fiber project since the beginning and knew all the complexities regarding the already existing data structure. Last but not least product owner in our scrum team provided me with all the important business knowledge and requirements needed to reach correct and practical results.

Assumptions and Limitations

One of the main assumptions for producing desired results is receiving accurate data from the business source system regarding the property owner's decision. There will be steps implemented to verify this belief, such as providing the data to teams with access to operational systems that show the real-life status of each address.

Another assumption is the willingness of the telecom company to share its data procedures, architecture design, and most importantly its data. After several discussions, it was

determined that all these factors can't be seen as a competitive advantage. The whole data journey is a pretty standard way of data processing, plus the whole IT sector is expected to migrate to the cloud and new technology stack, which will result in many changes. Moreover, the competitors already have significantly more data about connected buildings and households to organize their fiber deployment and sale of tariffs, so combined with the fact that the public release of this thesis will be delayed it won't bring them any additional value.

Despite all these facts I was advised to keep the name of the company anonymous to further enhance its safety. This is a limitation of the thesis since in the other scenario I could have included a concrete analysis of the company and its influence on the German market plus I would use more internal sources providing more specific info.

A further limitation is the obligatory anonymization of addresses to comply with GDPR rules. Also, I wasn't allowed to include any financial data, to prevent giving away any classified information.

Structure of the Thesis

The thesis has a structure according to the general recommendations of the Department of Information Technologies and Statistics at the Prague University of Economics and Business.

The structure of this thesis follows a well-organized format beginning with an Introduction, which provides an overview of the research topic and outlines the objectives, approach, and limitations of the study. Following the Introduction, the Literature Review delves into existing research and books related to the field of telecommunication companies, fiber optic infrastructure, and data processing providing a comprehensive analysis of relevant statistics, concepts, and findings. The subsequent chapter, Telecommunication Companies gives an overview of the field and focuses specifically on the technology of fiber optic and its predecessor copper wires, then it analyzes the situation of fiber rollout in Germany. The most crucial chapter called Data Journey explores different data layers, starting from the data acquisition layer, moving to the staging layer, core layer, data mart layer, and all the way up to the export layer. Each layer is examined from the theoretical perspective and then the implementation of concrete steps to process the property owner data comes in. The fourth chapter Visualization presents the findings and results derived from the data analysis, accompanied by meaningful insight. Finally, the Conclusion chapter synthesizes

the key findings, discusses their implications, and offers suggestions for future research directions.

Methodology

To achieve the goals of this thesis author used several methods, a comparison method to better understand the differences between copper and fiber technologies, then a literature review method, an observation method, and an analysis, for working with the data.

Expected benefit

The main benefit of the thesis from the perspective of the company is access to continuously updated data distinguishing between different types of landlord's decisions. This data is requested by several different teams and will help for example with an estimation of possible new tariff sales. Also, the previous message version 1 which has been used for the last couple of years will be cancelled in the future, and by integrating the new version 2 the team will be prepared in advance.

Another benefit lies in the thorough description of different data layers and our data structure. As the natural circulation in our team continues, new members have a tough time understanding the whole process and the significance of different steps in the data journey. This thesis can serve as a tool for onboarding. It provides an overview of the characteristics of each layer and doesn't just answer the "what" questions but also tries to explain why things are the way they are. This benefit has already been acknowledged by our newly appointed product owner who wants to be one of the first readers of the thesis.

Finally, the thesis can serve as a practical overview of the data engineering process for other data experts or beginners interested in the topic. It provides them with a basic understanding of layering, which is needed for designing new data structures. The implementation parts showcase some of the suggested approaches to take when encountering an issue. Given that I'm someone who entered the world of data just 2 years ago, the thesis is written in a language which should be understandable to most readers.

1 Literature Review

The following chapter outlines information sources, which were used in this thesis. It is not a complete list of cited sources, I picked the more fundamental and impactful ones. Books were the highest-represented category, but online sources and academic papers also played an important role in providing the theoretical foundation for the thesis.

1.1 Technical Literature

To gain at least a remote understanding of the actual cables and their different characteristics I used the book „Cabling: The Complete Guide to Copper and Fiber-Optic Networking”. It is a comprehensive book that covers the fundamentals and practical aspects of designing, installing, and maintaining cabling systems for both copper and fiber-optic networks. It explores many topics regarding cabling systems, but for the purpose of the thesis, I needed just general information about copper and fiber cables. (Oliviero & Woodward, 2014)

„Data Mining: Concepts and Techniques” is a foundational book that introduces readers to the principles and techniques of data mining. It covers various data mining concepts including preprocessing, classification, clustering, and more. It also distinguishes between different types of data, which provides some base for the subchapter about data variety. (Han et al., 2012)

„The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling” also addresses the importance of the staging area within the data warehouse architecture. The book in addition to others discusses how the staging area serves as a landing zone for raw data extracted from source systems before transformation and loading into another layer. It highlights strategies for structuring the staging area and handling data quality issues to ensure reliable and efficient data integration into the data warehouse. Information from this book ended up being used several times in the Staging Layer chapter, for example for data extraction and data warehouse classification. (Kimball & Ross, 2013)

„Designing data-intensive applications” by Martin Kleppmann is a book discussing many issues related to the topic of data in system design. One of the focuses is on processing and its different types. Choosing between full processing and delta processing is an important

decision data architects must make, and I used information from this book to explain some of the pros and cons of each approach. (Kleppmann, 2017)

„Database System Concepts” is a comprehensive textbook that covers fundamental principles and concepts related to database systems. One of the topics it covers is normalization which is seen as a key aspect of database design. It explains the process of organizing data into tables and applying normalization techniques to minimize redundancy and dependency issues. I focused specifically on the first three normal forms for my work, and this book provided valuable information on this topic. (Silberschatz et al., 2011)

William H. Inmon known as Bill Inmon is recognized as the father of the data warehouse, in this thesis I used a few of his books. Specifically, „Data Lake architecture: designing the data lake and avoiding the garbage dump”, „Data architecture: a primer for the data scientist”, and lastly „Building the data warehouse”. His expertise in the world of data helped me with a better understanding of different data structures and principles to follow when designing a new one. (Inmon, 2016)

„The Truthful Art: Data, Charts, and Maps for Communication” by Alberto Cairo is a highly regarded book that focuses on the principles of effective data visualization. Cairo emphasizes the importance of accuracy, clarity, and honesty in visualizing data. The book is filled with insightful examples and case studies that illustrate these principles, which helped me with the creation of the chapter about visualization. (Cairo, 2016)

1.2 Online Sources

The beginning of the thesis introduces readers to telecommunication companies and their endeavors. I used the internet encyclopedia Britannica to deepen my general understanding of some telecom services, which are described at the beginning of the thesis. The author used mainly segment „Television - History, Components, & Uses”. (Fink et al., 2024)

An article from sector strategist Slijkerman called „Startling differences in fibre connectivity across Europe”, showcases percentages and total numbers of connected households to fiber infrastructure across different European countries. It outlines the immense differences between some states but also displays the overall growing trend in this area. (Slijkerman, 2023)

A unique international forum OECD collects data from over 100 countries, and I used it to discover statistics about broadband subscriptions and different types of technology used. Another trustworthy, but more specialized organization is FTTH Council Europe. It is a

great source for statistics about fiber connectivity, but I didn't manage to find the most up to date 2024 numbers.

Enisa (European union agency for cybersecurity) published a document in 2022 „Deploying Pseudonymization Techniques – The case of the Health Sector”. This paper relatively simply demonstrates what pseudonymization is about and what are its use cases. I used this source to distinguish between anonymization and pseudonymization in the chapter Data Acquisition.

A blog post from Siva Ilango about the „Principles of Data layers in Data Platform” provides an overview of possible data layering architecture. It divides the process into 3 areas: The staging layer, the Core layer, and the Presentation layer, which is then divided between the Data mart section and the Report section. This is a very similar approach to the one I introduce in this thesis and therefore I was able to reuse some of the reasoning and characteristics of layers, especially the core layer. (Ilango, 2023)

The article „Data Mart Defined: What It Is, Types & How to Implement” by Andy Morris posted on the ORACLE NetSuite website provides general information about data marts and the differences between their types. I utilized mostly information about dependent and independent data marts. (Morris, 2021)

1.3 Academic Publications

To look for academic publications I mostly used the free web search engine Google Scholar. I also used search tools Semantic Scholar and CORE, which both provide millions of publicly available papers from all around the world.

„Comparative study between fiber optic and copper in communication link”, done by S. Babani, A. Bature, and M. I. Faruk, was utilized to bring some light on to differences between the legacy infrastructure and the new fiber optic cables. It compared attributes such as bandwidth, cost, weight, size, signal loss, and safety. (Babani et al., 2014)

A paper from the International Journal of Emerging Technology and Advanced Engineering called „Fiber optic communications: An overview” studies the development of communication via optical fibers. It outlines the advances in different fiber types, their properties, limitations, and applications. I used this research to confirm and extend the above-mentioned comparative study. (Sharma et al., 2013)

2 Telecommunication companies

Telecommunication companies serve multiple purposes in today's society. As the name suggests their original intent was to facilitate communication and while that still holds true means of fulfilling that purpose are continuously evolving.

The most well-known service provided by telecom companies is mobile phone connection. To make this possible a vast network infrastructure of cell towers and base stations had to be built, maintained, and optimized. So, when a user relocates from one location to another the network seamlessly hands off the communication with his phone to the tower with the strongest signal. As a result, the user can use phone services in the form of calling, sending, and receiving SMS and MMS messages, and perhaps even more importantly connecting to the internet.

Another well-established service is television connection, users can choose from Satellite TV, which provides high-quality video even in rural locations, classical Cable TV with reasonable pricing, or IPTV (Internet Protocol Television) with a flexible and more interactive user experience. (Fink et al., 2024)

Household internet connection is a third essential service offered by telecom companies. There are several ways to achieve a house connection, the main ones being copper cables, fiber, fixed wireless, DLS, satellite, and relatively new 5G. Factors customers have to consider when choosing the right option are internet speed, bandwidth, which means a rate of data transfer in a given amount of time, availability, price, and reliability.

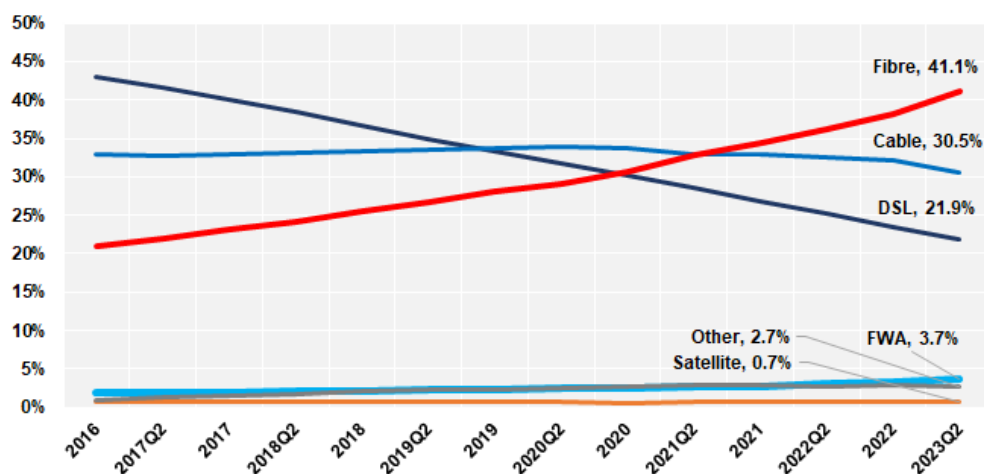


Figure 1 Subscriptions to broadband access technologies in OECD countries (OECD, 2023)

2.1 Copper

Wires made out of copper have been serving as transmitters of data since the invention of the telegraph in the 1830s. That kicked off the development of copper infrastructure. The telegraph was eventually replaced by the telephone, which was still dependent on copper. (Black, 1983) So the installation of more copper lines continued and later on in the 20. century, with the introduction of new coaxial cable and twisted pair cable, the copper infrastructure started getting used for television and internet connection as well.(Oliviero & Woodward, 2014)

Different types of copper cables have varying characteristics, for example, commonly used copper wire Cat 5e has a maximum bandwidth of 100Mbps while newer version UTP Cat 6 is capped at 1 Gbps. Cost is another variable with great differences, cat 5e is slightly less expensive than the fiber alternative, while Cat 6, Cat A6, and Cat 7 are in way higher price category. When it comes to latency copper-based Ethernet is limited by distance, cable length shouldn't exceed 100 m since the time it takes electrons to travel from one end of the cable to another would cause too big of a delay, plus there is a high transmission loss associated with longer lines. This is combated by installing a repeater or Ethernet switch.(Babani et al., 2014)

2.2 Fiber

The rollout of fiber is in most countries branded as an introduction to this new technology. That's why it might be surprising that the "fiber revolution" started already more than 40 years ago. In 1970 optical fiber was successfully developed by Corning Glass Works. Later on in the same decade came the first commercial fiber optic system, which operated at a bit rate of 45 Mbps, and its repeaters for signal amplification were spacing of up to 10 km. (Patibandla et al., 2020) From that point on countries have been slowly shifting from copper to fiber. The process has been speeding up in the last 10 years, some countries have already exceeded 80 % of connected homes, while others like UK, Germany, or Belgium are still dependent on the legacy infrastructure (copper-based). (Slijkerman, 2023)

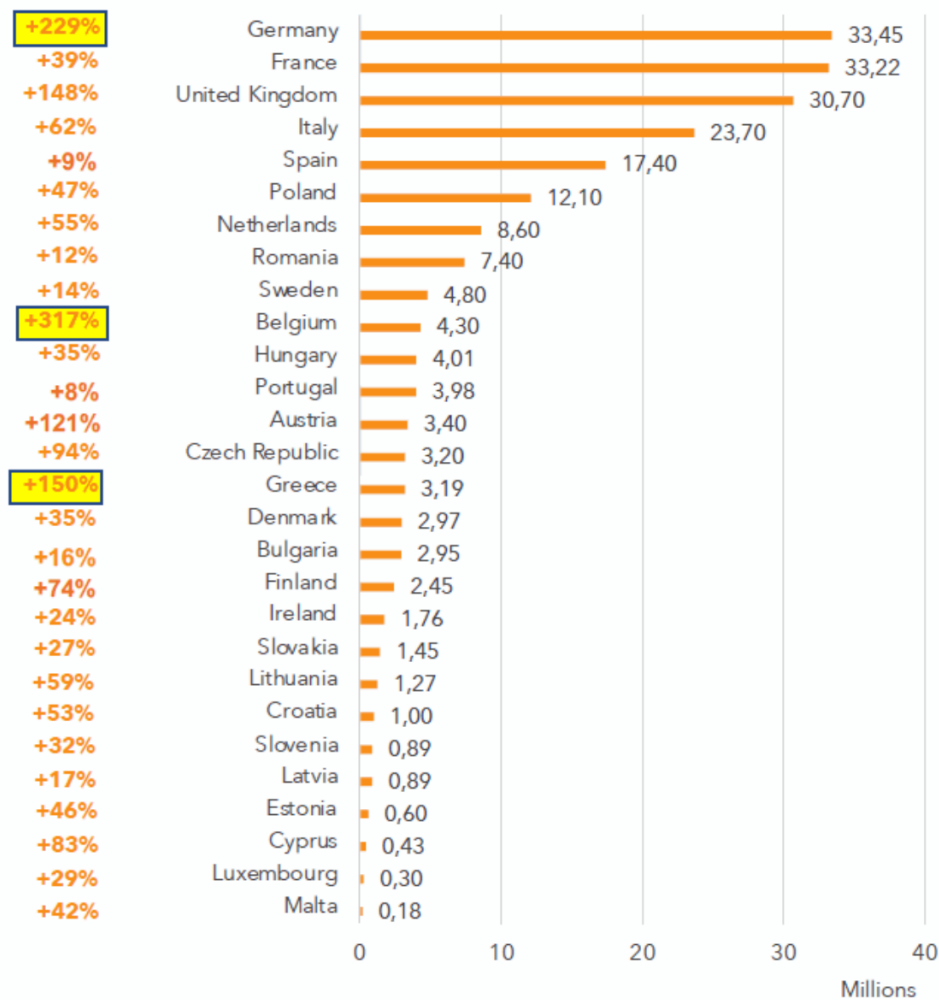


Figure 2 Forecast for FTTH/B passed in EU from 2023-2028 (Jackson, 2023)

Fiber optic cable broadband is continuously evolving. Achievable bandwidth is nowadays above 400 Gbps and researchers are expecting an increase to Tbps in the future. (ITU/UNESCO Broadband Commission for Sustainable Development, 2021) Another feature of optical fibers is their low transmission losses even with greater distances, this means the cost of additional equipment such as repeaters is reduced. Compared to the 10 km in the first generation of fiber optic systems, today's technology is capable of safely transmitting data over 100 km without a repeater. (Sharma et al., 2013)

Latency is another important topic in the subject of data transmission and fiber is known for its low-latency characteristics. It's because it doesn't rely on electrons to deliver the signal, but it uses light. The speed of light in a vacuum or air is 300 000 km/s and inside the cable it's capable of traveling about 2/3 of this speed, it's slowed down a little due to the refractive index of the glass or plastic core. (The Fiber Optic Association, 2023)

Fiber optic cables have a couple of other beneficial traits, one of them being their small size and low weight, especially compared to heavy and bulky copper cables. They are also safer

because of the absence of ground loops, sparks, and high voltages. The only concern could be eye damage due to laser light emissions. Last but not least any signals that might escape are absorbed by a protective coating. This makes it more secure than copper wires, where electrical signals can be easily intercepted. (Sharma et al., 2013)

2.3 Fiber in Germany

Germany has struggled to keep pace with other European countries in the deployment of fiber optic infrastructure. While some countries have close to 90 % of houses connected, Germany is only at a little over 22 %, leaving more than 32 million households without fiber connection. (FTTH Council Europe, 2022) As we can see in Figure 1 there are states with less robust economies with far greater results. This is partially caused by significant differences in the priority governments assigned to FTTH in their countries. For example, Spain saw the fiber roll-out as a way to retain the population, especially younger generations, in the depopulated countryside and therefore allocated a lot of resources to support companies connecting even distant and small villages. (Merino et al., 2024)

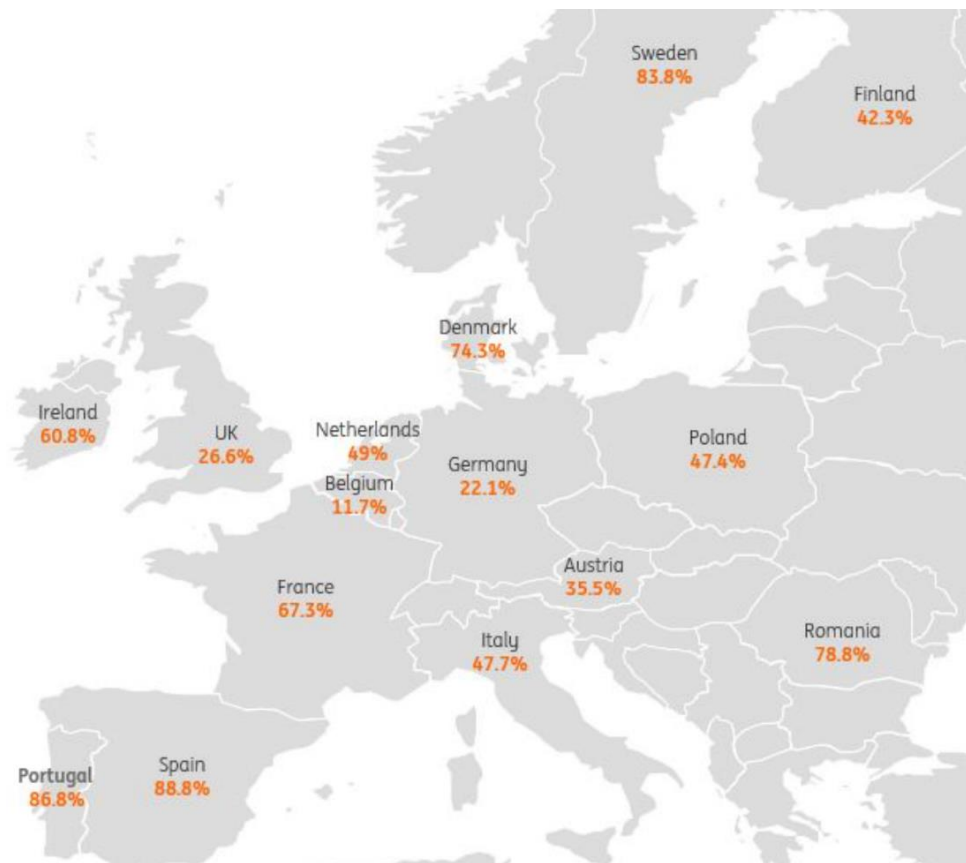


Figure 3 Percentage of households with fiber connectivity in Europe 2022 (FTTH Council Europe, 2022)

One of the other reasons for Germany's slow transition to fiber is low demand from customers. Data from 2018 shows that 81 % of private households have at least 30 Mbps available and 65 % of households have at least 100 Mbps. These numbers aren't outstanding, but they do make Germany slightly above average compared to other EU27 members, and more importantly, it means that most of the population can be supplied with enough broadband speed for normal usage. Combine this information with an estimated cost for nationwide fiber-optic expansion being well over 30 billion euros and it's clear why hasn't Germany been one of the frontrunners in fiber deployment. (Girard, Mattes and Michelsen, 2018)

Despite these circumstances, Germany realized that to stay globally competitive and meet the future needs of their citizens fiber optic infrastructure is going to be a necessity. Also, the general consensus is that just like copper, fiber infrastructure will be sufficient for at least another 40 years and therefore it's reasonable to invest in it. To achieve this German federal government introduced the Gigabit Strategy in July 2022. The ultimate goal of this incentive is to bring FTTH to every household in Germany by 2030 and the first step to get there is connection of 50 % of households by 2025. (Press and Information Office of the Federal Government, 2023)

2.4 Chapter conclusion

This chapter provided readers with a basic understanding of the main services offered by telecommunication companies. The focus was on two types of infrastructures used for internet connection, the original one made out of copper cables and the state-of-the-art one made out of fiber optic cables. Other than their history it compared some of their characteristics such as bandwidth, distance without repeaters, safety, latency, and more. From this analysis, it was clear that fiber is superior in every domain, and as the history of development in this area suggests it will become the dominant infrastructure for the next several decades.

Furthermore, the chapter analyzed the current situation of FTTH in Germany. Compared the percentage of households connected to other European countries and suggested some reasons for their relatively low results. It also revealed Germany's reaction, which is an ambitious goal for nationwide fiber optics coverage by 2030. To get there Germany needs to spend billions of euros every year. Implementing such a project requires an organization guided by useful and factually correct data, how to achieve this state, will be explored in the next chapter.

3 Data Journey from Acquisition to Insights

In this chapter, readers embark on a systematic exploration of the data journey, traveling through its distinct layers from acquisition to insights. Beginning with data acquisition, they explore the complexities of sourcing and collecting diverse data types, including structured and unstructured data. Then they progress to the stage layer, where data undergoes initial processing and storage. Proceeding into the core layer, readers dive into the heart of data processing, where advanced analytics and modeling techniques are applied to uncover hidden patterns and trends. Next, they navigate through the data mart layer, where subsets of data are tailored to meet specific business needs and facilitate targeted analysis. Finally, the journey concludes at the export layer, where insights are extracted and presented in accessible formats for consumption by stakeholders. After analyzing each layer, readers will look at the specific technology, code, and actions that the author has taken to build up the processing for data regarding the landlord's decision. Throughout this expedition, the critical role of each layer in the data journey is displayed, emphasizing the importance of seamless integration and attention to detail for unlocking actionable insights and supporting data-driven decision-making.

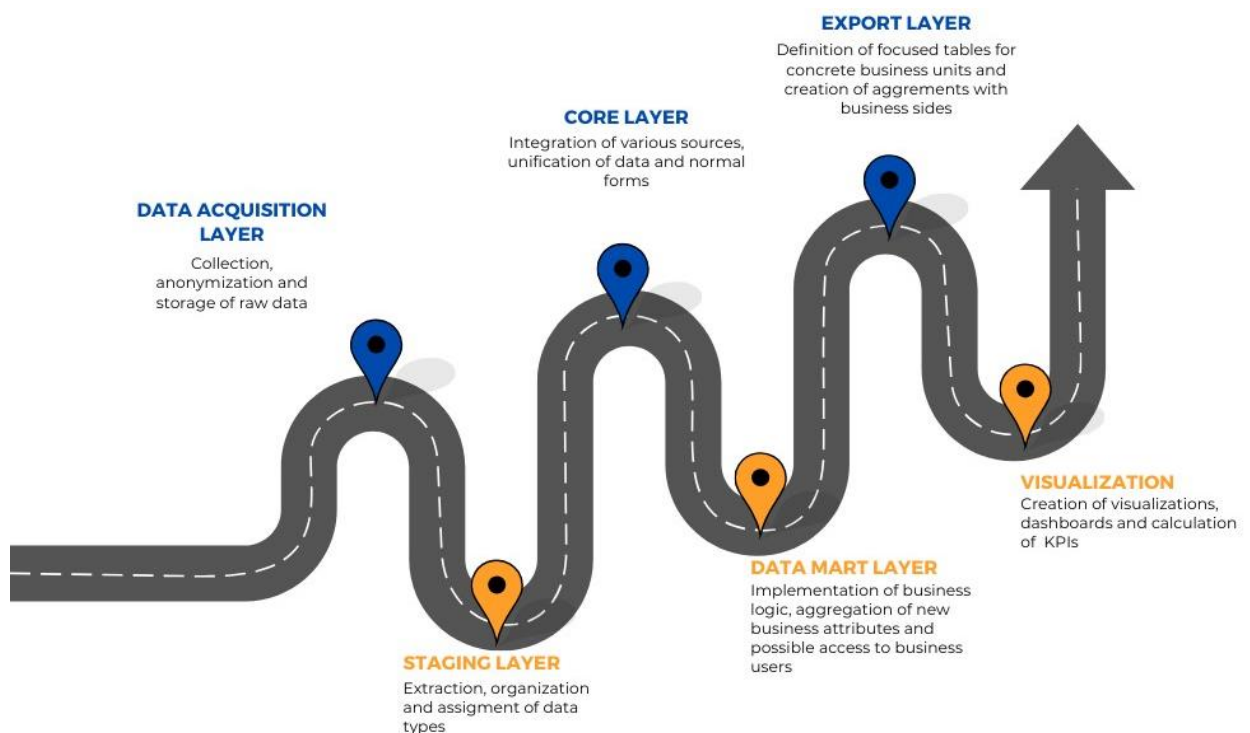


Figure 4 Data Journey visualization on the roadmap (source: author)

3.1 Data Acquisition Layer

3.1.1 Broad overview

Data acquisition forms the bedrock of any data-driven endeavor. It encompasses the systematic collection, processing, and storage of raw data from diverse sources, ranging from traditional databases to digital interfaces and sensor networks. Technical experts design and implement data acquisition pipelines, ensuring the seamless integration of diverse data streams while maintaining reliability and efficiency. This process demands expertise in various domains, including signal processing and data storage architectures. (Di Paolo Emilio, 2013)

Engineers leverage advanced technologies such as real-time streaming and distributed computing to handle the volume and velocity of data generated in today's digital landscape. Each component, from data capture devices to backend databases, must be carefully engineered to ensure data integrity, scalability, and compatibility with downstream analytics and applications. Data acquisition serves as the foundational step in the data journey, and its accurate execution is essential for the following analyses and insights to be correct and meaningful.

3.1.2 Data Variety

When considering the collection of different types of data, the distinction between structured and unstructured data becomes crucial. Structured data are characterized by their organized format with well-defined schemas, providing information in the form of predefined attributes such as name, date, price, or address. Typically sourced from databases, spreadsheets, or structured APIs, acquiring structured data involves establishing efficient pipelines for extraction and loading. This could be done in arranged intervals for example on an hourly or daily basis, or if the application requires it, data can be collected using real-time streaming. (Han et al., 2012)

On the other hand, the collection of unstructured data demands a more complex approach. Unstructured data, which encompasses diverse formats such as text, images, and sensor readings, requires specialized techniques for acquisition. Data collection from unstructured sources often involves methods like web scraping, natural language processing, or image recognition to capture and transform disparate data into a standardized format suitable for analysis and storage. (Han et al., 2012)

The third option is semi-structured data which is a combination of both structured and unstructured data and has some characteristics of each one of them. It does not conform to the rigid structure of traditional relational databases but still exhibits some level of organization. Common examples of semi-structured data include XML files, JSON documents, and NoSQL databases. While semi-structured data may contain nested or hierarchical elements, it does not require a fixed schema, allowing for flexibility in data representation and storage. This characteristic makes semi-structured data well-suited for handling diverse data types and accommodating evolving data requirements, making it increasingly common in modern data-driven applications and systems. (Lin et al., 2018)

3.1.3 Anonymization and pseudonymization

Anonymization and pseudonymization represent critical initial steps when collecting data, particularly in protecting privacy and complying with data protection regulations. Anonymization involves the irreversible process of removing or encrypting personally identifiable information from datasets, making individuals unidentifiable. This ensures that even with access to the data, individuals cannot be directly identified. Pseudonymization, on the other hand, replaces identifiable information with pseudonyms or aliases, allowing for data processing without revealing the actual identities of individuals. Both techniques serve to mitigate privacy risks while still enabling useful analysis and research. By implementing these measures early in the data collection process, organizations can uphold privacy standards and reduce the risk of data breaches. (Enisa, 2022)

3.1.4 Implementation

In the scenario of this thesis, all the data being collected comes from an operational business system in a structured or semi-structured format. This makes the step of data acquisition relatively simple as the data collection and anonymization is done by Acl (acquisition layer), which is a layer between the operational system and the dispositive system (my team). To complete the task, I reached out to the Acl team via email, requesting a subscription to the latest version (version 2) of "DigiOSS – BuildUpAgreementEvent". This updated version includes data regarding the distinction of property owner decisions. Alongside the subscription request, I submitted an XML file containing detailed instructions for anonymizing and pseudonymizing specific attributes. In order to correctly add the structure and attributes of the new message type into the XML file, I needed Swagger files provided by Acl. These Swagger files offer information on the structure of all messages originating from the source system.

In the pictures below we can see some snippets of the XML file, displaying the demand for pseudonymization of attributes like klsId or fiberOnLocationId, because they can be used to track down the address of the customer. Then attribute locationSpecifics must be anonymized since it could contain sensitive information. But the absolute majority of other attributes can be kept in their original form.

```

▼<EinAttribut>
  <Attribut>locationSpecifics</Attribut>
  ▼<AttributBeschreibung>
    <![CDATA[ It's in hand of consumers. First UC is implemented by exploration-app and ibt-core
  </AttributBeschreibung>
    <Typ>AlphanumerischVariabel</Typ>
    <Laenge/>
    <PersBez>Kunde</PersBez>
    <PersBezBI>personenbeziehbar</PersBezBI>
    <AnoPseudo>anonymisiert</AnoPseudo>
    <APTransformation>NULL</APTransformation>
  </EinAttribut>
</Pfad>
</ObjektListe>
</SchnittstellenDatenKomponente>

```

Figure 5 Anonymization of "locationSpecifics" (source: author)

```

<!-- Eingehende Assoziation: BuildUpAgreementEvent_eventPayload -->
▼<Pfad>
  <JsonPfad>$.eventPayload</JsonPfad>
  ▼<Objektbeschreibung>
    <![CDATA[ The eventPayload contains a reference to the building part
  </Objektbeschreibung>
  ▼<EinAttribut>
    <Attribut>klsId</Attribut>
    ▼<AttributBeschreibung>
      <![CDATA[ Kls id of the installation address. ]]>
    </AttributBeschreibung>
    <Typ>Ganzzahl</Typ>
    <Laenge/>
    <PersBez>Kunde</PersBez>
    <PersBezBI>personenbeziehbar</PersBezBI>
    <AnoPseudo>pseudonymisiert</AnoPseudo>
    <APTransformation>Bdm_KLS_id_PS</APTransformation>
  </EinAttribut>
  ▼<EinAttribut>
    <Attribut>fiberOnLocationId</Attribut>
    ▼<AttributBeschreibung>
      <![CDATA[ Synthetic key of distribution point. ]]>
    </AttributBeschreibung>
    <Typ>AlphanumerischVariabel</Typ>
    <Laenge/>
    <PersBez>Kunde</PersBez>
    <PersBezBI>personenbeziehbar</PersBezBI>
    <AnoPseudo>pseudonymisiert</AnoPseudo>
    <APTransformation>Bdm_Technische_ID_PS</APTransformation>
  </EinAttribut>

```

Figure 6 Pseudonymization of "klsId" and "fiberOnLocationId" (source: author)

```

▼<EinAttribut>
  <Attribut>agreementStatusNE3</Attribut>
  ▼<AttributBeschreibung>
    <![CDATA[ Characteristics: [APPROVED | REJECTED | WAITING | POSTPONED] WAITING
      the build-up decision. The customer journey does not go any further. ]]>
    </AttributBeschreibung>
    <Typ>AlphanumerischVariabel</Typ>
    <Laenge/>
    <PersBez>-</PersBez>
    <PersBezBI>-</PersBezBI>
    <AnoPseudo>-</AnoPseudo>
    <APTransformation>-</APTransformation>
  </EinAttribut>

```

Figure 7 Request for an unmodified attribute in XML file (source: author)

After successfully completing the communication process, I awaited the monthly Acl deployment before receiving the BuildUpAgreement version 2 data. Upon its arrival, the data was stored in ODL (One Data Lake), facilitating direct accessibility for our team. From that point on, the process follows the standard procedure, with Acl collecting new data every 5 minutes and transmitting it to ODL for storage and accessibility.

3.2 Stage Layer

3.2.1 Broad Overview

The staging data layer is like a middleman between raw data sources and where data is stored for analysis. It's where unpacking, organization, and parts of the preparation of the data take place before proceeding to the next step. This layer helps to make sure the data is in the desired format, for example, dates are displayed with time zones being accounted for and with the same granularity. Another task being done at this stage is assigning correct data types to attributes.

An illustration to better imagine the purpose of the stage layer could be one of a professional kitchen. The food arrives from different locations in big boxes and first, it must be unpacked and organized into its designed storage space. Some of the ingredients might need to be precut or preprepared and some ingredients are labeled with names and dates of expiration/arrival. During this whole process, only professional staff can access the food, and entrance to the kitchen is prohibited to anyone else, just like it's supposed to be with the stage layer, which is not the place for any requests from the customer side. (Kimball & Ross, 2013)

3.2.2 Data extraction

Data extraction is the first step in the staging layer, it's the pivotal process of retrieving data from its source locations. It involves accessing and collecting data from diverse sources such as databases, files, or ODL. In the context of the staging layer, data extraction often requires taking apart and interpreting the raw data format, whether it be structured like a relational database or semi-structured like JSON or XML. (Kimball & Ross, 2013)

For JSON, this often involves using programming languages like Python or JavaScript, with techniques like JSONPath. Alternatively, databases like PostgreSQL, MySQL, and SQL Server offer SQL support for querying JSON data directly within SQL statements using specialized functions and operators. Similarly, for XML, libraries such as lxml in Python can be used to extract data by navigating the XML document's tree structure and selecting specific elements or attributes. (Klessinger et al., 2023)

Overall, effective data extraction is fundamental to the success of the data integration process, as it sets the stage for subsequent data transformation activities.

3.2.3 Full processing and Delta processing

There are two primary processing methods commonly employed in the staging layer: full processing and delta processing. Full processing involves extracting and transforming all available data from the source into the staging area. This approach ensures that the staging layer contains a complete snapshot of the source data at a specific point in time. By including all available data, it eliminates the risk of missing any important information or updates. Additionally, this approach simplifies the development and adjustment of code. During deployment, developers can easily drop the existing table and create a new one with an updated schema, without the need to migrate previous data. Overall, full processing offers a robust and reliable method for ingesting and managing data in the staging layer, but it can be highly demanding in terms of resources and time. (Kleppmann, 2017)

In contrast, delta processing focuses on capturing only the changes or updates that have occurred since the last processing cycle. By identifying and processing only the delta, or incremental changes, delta processing minimizes the amount of data transferred and processed, reducing processing time and resource consumption. This approach is particularly helpful in projects that gather vast amounts of data daily, ranging from hundreds of gigabytes to terabytes. Similarly, it proves invaluable for projects that have been in operation for some time and have accumulated loads of data over time. On the other hand, delta processing presents challenges for developers due to the complexity involved in

migrating data, often requiring complicated data migration scripts and thorough testing procedures. (Akidau et al., 2018)

Both full processing and delta processing play crucial roles in data integration pipelines, with full processing ensuring data completeness and delta processing enhancing efficiency and scalability by handling only the incremental changes. Balancing between these two approaches based on specific use cases and requirements is key to optimizing the performance and effectiveness of the staging layer in the data integration process.

3.2.4 Storage

There are various options for storing data in the staging layer. Some of them are for example NoSQL databases, cloud storage services, and file systems. This subsection will focus on a few of the most commonly used storage solutions, discussing their functionalities and how they contribute to efficient data management in the staging environment. It'll be a data warehouse, relational database, and data lake.

Data Warehouse

A data warehouse serves as a centralized repository for storing structured, integrated, and historical data, making it a cornerstone in the staging layer of a data architecture. Data warehouses are designed to support analytical processing, facilitating complex queries and reporting for decision-making purposes. Unlike traditional relational databases, data warehouses typically follow a dimensional modeling approach, organizing data into fact tables and dimension tables. This dimensional model provides a logical structure that aligns with business reporting needs, allowing for efficient querying and analysis. While data warehouses excel at handling structured data and supporting complex analytics, they may require significant upfront planning and investment in terms of design and maintenance. (Kimball & Ross, 2013)

Relational Database

Relational databases are another common choice for storing data in the staging layer, offering flexibility, and robust transactional support. Unlike data warehouses, which are optimized for analytical processing, relational databases are well-suited for transactional processing and support a wide range of data types and relationships. They provide ACID (Atomicity, Consistency, Isolation, Durability) compliance, ensuring data integrity and reliability. Relational databases offer a familiar SQL-based interface for data manipulation and querying, making them accessible to a wide range of users. However, relational

databases may face scalability challenges when dealing with large volumes of data, and their schema-on-write approach may require frequent updates as data sources evolve. (Garcia-Molina et al., 2009)

Data Lake

Data lakes have emerged as a popular choice for storing diverse and unstructured data in the staging layer, offering scalability, flexibility, and cost-effectiveness. Unlike traditional data warehouses and relational databases, which impose structure upfront, data lakes embrace a schema-on-read approach, allowing data to be ingested in its raw form and structured as needed during analysis. This flexibility makes data lakes ideal for storing large volumes of raw data, including semi-structured and unstructured data such as log files, sensor data, and social media feeds. However, managing data quality, governance, and security in a data lake environment can be challenging, requiring robust metadata management and data governance frameworks. Additionally, data lakes may require specialized skills and tools for data processing and analysis, since the sheer amount and variety of the data could easily turn it into a “one-way” data lake, which ends up being just relatively cheap storage of useless data. (Inmon, 2016)

3.2.5 Technology Stack Overview

In this subsection author presents concrete tools used for the implementation part of this thesis. Most of the technologies and tools mentioned below are not unique to the staging layer, the process of developing, testing, and deploying new code is similar to staging, core, data mart, and export layer.

ODL

In the staging layer, as well as throughout other layers of this project, we utilize ODL for storing our data, which is provided by the American software corporation Cloudera. ODL serves as a massive data lake shared by multiple teams within the company, offering fixed resources that are distributed among various data engineers. However, due to the high demand, especially during peak hours, the platform can experience slowdowns, that's one of the reasons why the company decided to migrate to the Google Cloud platform, which takes place in 2024.

PRD and REF

Part of ODL are different environments with different use cases. In my team, we use PRD (production environment) and REF (reference environment). REF's purpose is to provide

safe and efficient conditions for developing and testing the code. It achieves that by providing just a small percentage of all of the data, which results in quicker processing, and if the developer writes flawed code or loses any data it doesn't do any harm, since no one relies on any results coming from REF. PRD is a different story since that's where the up-to-date tables with all the data are stored. Mistakes in PRD carry significant consequences, potentially leading to export delays or outdated reports, this proves the importance of thorough testing on REF beforehand.

Apache Hive

One unique characteristic of the staging layer in this project is the use of Apache Hive. This distributed and fault-tolerant data warehouse system is used for processing and analyzing data. Built on top of Apache Hadoop, Hive facilitates the writing, reading, and management of petabytes of data via SQL¹, providing a robust framework for our data processing needs. In addition, compared to Apache Impala, which is used throughout the rest of the project. It provides some functions necessary for extracting information from JSON files and is highly reliable, both crucial attributes for the initial processing stages. The main downside of Apache Hive is the slower speed compared to Impala, that's why it's only used where necessary.

DBeaver

For executing SQL queries and testing I used DBeaver PRO, a *database tool for developers, database administrators, analysts, and everyone working with data. It supports all popular SQL databases like MySQL, MariaDB, PostgreSQL, SQLite, Apache Family, and more.*² For the staging layer, I set up a connection via Hive to access the ODL cluster.

Visual Studio Code and Gitlab

Following the development and testing phases, the code is rewritten into Visual Studio Code, which is a source code editor developed by Microsoft. That's where changes can be committed and merged, initially into the Develop branch which is then merged into the Production branch. This process of storage, version control, and deployment starts in Visual Studio Code but is mainly done in GitLab.

¹ Parts of the definition of Apache Hive were taken from the official website <https://hive.apache.org/>

² Definition was taken from official website <https://dbeaver.io/>

KNIME

Merge in GitLab starts a pipeline which if run successfully produces an artifact in the form of a file containing KNIME workflow. KNIME is another tool used in this project. KNIME offers a complete platform for end-to-end data science, from creating analytic models to deploying them and sharing insights within the organization, through data apps and services.³ In our scenario KNIME is used for automating the daily processing of the data, it lets users create different types of nodes, with executable code inside, and these nodes are organized into a workflow. This workflow is then scheduled for nighttime, when it automatically starts, and in the morning developers either get updated, processed data, or error log messages.

JSON Editor online

To correctly unpack JSON files, one must locate specific attributes within the data. To ease the search inside the long lines of every type of JSON, I used the basic JSON Editor available online. Its main function is to transform the provided JSON and display it in its hierarchical structure, which is significantly easier to navigate.

3.2.6 Implementation

The goal of this layer is to extract all the new information out of the JSONs and incorporate it into the already existing object `CL_TMagic_StageBuildUpAgreementEvent_MT`. This object already stages “DigiOSS – BuildUpAgreementEvent” version 1, and a couple of other historic message types. That makes the task of staging more complex since the idea behind the stage layer is to extract and process the information in a way that users at the following layers won’t have to differentiate between message types and versions.

Two of the most crucial new attributes are `agreementStatusNE3`, showing the landlord's decision on installing GFTA one box in the basement, and `agreementStatusNE4`, displaying his reaction to installing a fiber socket to the concrete apartment. These attributes can hold one of 4 values “APPROVED”, “REJECTED”, “WAITING” or “POSTPONED”. Now the task is to stage these attributes and a special requirement from the product owner is to take the property owner's decision from historic message types and versions and interpret it as his decision for both NE3 and NE4.

³ Description taken from official website <https://www.knime.com>

The difference is that previously property owner decision was stored in an attribute called eventType, which can hold many other unrelated values and is missing the status “POSTPONED”. That will pose more challenges down the road, but for the staging layer, a simple CASE command does the job.

```
NVL(GET_JSON_OBJECT(JsonStruct, '$.eventPayload.agreementStatusNE3'),
CASE eventType
WHEN 'PROPERTYOWNER_CONFIRMATION' THEN 'APPROVED'
WHEN 'PROPERTYOWNER_REJECTION' THEN 'REJECTED'
WHEN 'PROPERTYOWNER_WAITING' THEN 'WAITING'
ELSE NULL
END) AS agreementStatusNE3,
NVL(GET_JSON_OBJECT(JsonStruct, '$.eventPayload.agreementStatusNE4'),
CASE eventType
WHEN 'PROPERTYOWNER_CONFIRMATION' THEN 'APPROVED'
WHEN 'PROPERTYOWNER_REJECTION' THEN 'REJECTED'
WHEN 'PROPERTYOWNER_WAITING' THEN 'WAITING'
ELSE NULL
END) AS agreementStatusNE4,
```

Figure 8 Staging of agreementStatusNE3 and agreementStatusNE4 (source: author)

The next attributes that need to be staged are the dates of the property owner's decisions. In the BuildUpAgreement v2, dates are stored at '\$.eventPayload.agreementDateNE3' (NE4 is similar). Unfortunately, in the other message types and versions, the location of the data is often different. To prevent having to write unreadable, repetitive code which would take about 7 long lines per date. I created a WITH clause with artificial attribute date_location, this table is connected via LEFT OUTER JOIN to the main table on attributes message type and message version. Then instead of staging every historic message type separately, I installed date_location into the GET_JSON_OBJECT command and it points to the correct location for each type and version.

```
WITH date_location AS
[
SELECT 'IBT - KLSEvent' AS message_type,
'1' AS message_version, '$.eventPayload.modificationDate' AS date_location
UNION ALL SELECT 'IBT - KLSEvent' AS message_type,
'2' AS message_version, '$.eventPayload.dateInformation.date' AS date_location
UNION ALL SELECT 'IBT - BuildingPartEvent' AS message_type,
'1' AS message_version, '$.eventPayload.dateInformation.date' AS date_location
UNION ALL SELECT 'DigioSS - BuildUpAgreementEvent' AS message_type,
'1' AS message_version, '$.eventPayload.agreementDate' AS date_location
], max_loadnumber_processed
```

Figure 9 WITH clause for the location of the date (source: author)

Since I worked with dates, I needed to cast them as timestamps and take care of the time zones as well.

```
IF (eventtype IN ('PROPERTYOWNER_CONFIRMATION', 'PROPERTYOWNER_REJECTION', 'PROPERTYOWNER_WAITING'),
    FROM_UTC_TIMESTAMP(CAST(REGEXP_REPLACE(GET_JSON_OBJECT(JsonStruct, 'date_location'), '[A-Z]', ''))
    AS TIMESTAMP), 'Europe/Berlin'),
    --- 'DigiOSS - BuildUpAgreementEvent' v2 doesn't have any of the above mentioned eventtypes
    FROM_UTC_TIMESTAMP(CAST(REGEXP_REPLACE(GET_JSON_OBJECT(JsonStruct, '$.eventPayload.agreementDateNE3'),
    '[A-Z]', '')) AS TIMESTAMP), 'Europe/Berlin')) ..... AS agreementDateNE3,
```

Figure 10 Staging of agreementDateNE3 (NE4 is similar) (source: author)

The location of other new attributes was found using JSON Editor and then they were staged simply using the GET_JSON_OBJECT command. Some of the new attributes might not end up being used, but the general rule is to stage them anyway because if they are needed in the future, they will be available without the need for time-consuming redeployment of staging.

```
GET_JSON_OBJECT(JsonStruct, '$.eventPayload.expectedAgreementDate') ..... AS expectedAgreementDate,
GET_JSON_OBJECT(JsonStruct, '$.eventPayload.expectedAgreementDateIsCalculated') ..... AS expectedAgreementDateIsCalculated,
GET_JSON_OBJECT(JsonStruct, '$.eventPayload.locationSpecifics') ..... AS locationSpecifics,
```

Figure 11 Other new attributes being staged (source: author)

The WHERE section of the object is used for setting limiting conditions. It displays the periods when distinct message types and versions were used. During development, I found some version inaccuracies which I fixed, and then added limiting acl_dop, which has a date of processing in it, to BuildUpAgreement version 1. Finally, I created a new condition for BuildUpAgreement version 2 and it's acl_dop starting 25.11.2023, the date when we started receiving this new version.

```
AND
-- 1. consider all Messages KLSEvent V1 until 2020-01-31
--- (messagetype = 'IBT - KLSEvent' AND messageversion = '1' AND acl_dop < '20200201000000')
-- 2. consider all Messages KLSEvent V2 starting from 2020-02-01 until 2020-07-31
--- OR (messagetype = 'IBT - KLSEvent' AND messageversion = '2' AND acl_dop >= '20200201000000' AND acl_dop < '20200801000000')
-- 3. consider all BuildingPartEvents (V1) starting from 2020-08-01, but the PROPERTYOWNER_[CONFIRMATION|REJECTION] are missing
--- OR (messagetype = 'IBT - BuildingPartEvent' AND messageversion = '1' AND acl_dop >= '20200801000000')
-- 4. consider all KLEvents (V2) of type 'PROPERTYOWNER_CONFIRMATION' AND 'PROPERTYOWNER_REJECTION' starting from 2020-08-01
--- OR (messagetype = 'IBT - KLSEvent' AND messageversion = '2' AND acl_dop >= '20200801000000' AND acl_dop < '20201106085109'
--- AND eventType IN ('PROPERTYOWNER_CONFIRMATION', 'PROPERTYOWNER_REJECTION'))
-- 4b. e-mail to propertyOwner for exploration_date_planned; will come within since '20201106102131'
--- eventtype EXPLORATION_DATE_PLANNED from case 3
--- OR (messagetype = 'IBT - KLSEvent' AND acl_dop >= '20200801000000' AND acl_dop < '20201106102131' AND
--- jsonStruct LIKE '%OWNER_EXPLORATION_DATE_APPOINTMENT_REQUEST%')
-- 5. consider the BuildUpAgreementEvent V1 since 2020-11-06 until 2023-11-25
--- OR (messagetype = 'DigiOSS - BuildUpAgreementEvent' AND messageversion = '1' AND acl_dop >= '20201106085109'
--- AND acl_dop < '20231125233752')
-- 6. consider the BuildUpAgreementEvent V2 since 2023-11-25
--- OR (messagetype = 'DigiOSS - BuildUpAgreementEvent' AND messageversion = '2' AND acl_dop >= '20231125233752')
```

Figure 12 WHERE conditions to manage types and version (source: author)

After the code was tested in DBeaver I ran the updated workflow in KNIME on REF for further confirmation that the changes won't cause issues to other operations. This went successfully so I proceeded with the deployment. The staging layer is the only layer in our

project that utilizes delta processing in its daily run, this is caused by the use of Apache Hive and its lack of speed. But when it comes to deployment of staging there is an exception, as we use the standard procedure of dropping the affected table and creating a new one from scratch, to avoid complicated migration of historic data.

3.3 Core Layer

3.3.1 Broad Overview

In data architecture, the core layer functions as a vital transition point, facilitating the seamless integration of data from various sources into standardized entities. This layer operates as an integration hub, where disparate data attributes are identified, mapped to consistent names, and unified into a cohesive model. By identifying business keys and applying universal cleansing rules, the core layer ensures data uniformity and integrity across the organization's ecosystem. (Inmon et al., 2019) Much like a diligent kitchen chef who prepares ingredients for cooking, the core layer diligently organizes the data preparation process, similar to cutting vegetables, peeling apples, and grating cheese, laying the groundwork for subsequent data processing and analytics tasks. (Ilango, 2023)

3.3.2 Data Unification

Unifying names for attributes from different sources is a crucial aspect of data integration, especially within the data core layer of an organization's architecture. This process involves aligning attribute names across disparate data sets according to the organization's naming conventions. Often, companies have varied naming conventions across different systems or departments, leading to inconsistencies. Additionally, the values within attributes may require standardization into a uniform format for accurate analysis. By unifying attribute names and values, data analysts and business users can easily understand and work with data from multiple sources, enhancing effective data integration and querying and facilitating more accurate analysis and reporting. (Holzapfel, 2023)

3.3.3 Normal Forms

Normalization is a crucial process within the data core layer of an organization's architecture, aimed at optimizing data storage, reducing redundancy, and enhancing data integrity. It involves organizing data into well-structured tables, ensuring that each table represents a single subject and that data dependencies are minimized. By adhering to

normalization principles, data engineers can effectively manage complex data sets, streamline data retrieval, and support efficient querying and analysis. (Silberschatz et al., 2011)

1NF

First Normal Form (1NF) is the initial stage of normalization, focusing on eliminating repeating groups and ensuring atomicity within each table. In the context of the data core layer, 1NF means ensuring that each attribute contains only atomic values and that there are no repeating groups or arrays. (Silberschatz et al., 2011) By restructuring data into individual columns with single values, 1NF simplifies data manipulation and enhances data consistency, laying the groundwork for further normalization. (Peterson, 2024)

2NF

The second Normal Form (2NF) builds upon the foundation established by 1NF, addressing partial dependencies within the data schema. In the data core layer, achieving 2NF involves ensuring that each non-key attribute is fully functionally dependent on the entire primary key. This means decomposing tables and organizing data to eliminate redundant information. By resolving partial dependencies, 2NF enhances data integrity and facilitates more efficient data management and querying processes. (Kolade, 2022)

3NF

The third Normal Form (3NF) represents a higher level of normalization, aimed at eliminating transitive dependencies within the data schema, while also following 2NF principles. In the data core layer, achieving 3NF involves ensuring that each non-key attribute is non-transitively dependent on the primary key. This requires further decomposition of tables and organizing data to remove indirect relationships between attributes. By eliminating transitive dependencies, 3NF reduces data redundancy, improves data integrity, and promotes a more streamlined and efficient data architecture. (Silberschatz et al., 2011)

In most data projects, the normalization process typically aims to achieve at least the Third Normal Form (3NF). But, in some cases, data projects may go beyond 3NF and aim for higher levels of normalization, such as Boyce-Codd Normal Form (BCNF) or even Fourth Normal Form (4NF) or Fifth Normal Form (5NF), if deemed necessary for the project's objectives. However, higher levels of normalization can result in more complex database schemas and queries, which may impact performance and require careful consideration. (Peterson, 2024)

3.3.4 Implementation

Technology stack

The tools used in the core layer are almost entirely identical to the ones used in the previous staging layer. The main exception is a switch of Apache Hive for Apache Impala. As I no longer have to extract information from JSONs or use some of the other features unique to Hive, Impala is a superior option mainly because of the significantly higher processing speed.

Initial processing

The first step of processing property owner decisions in the core layer is to update the already existing object `CL_TMagic_BuildUpAgreementEvent_MT`. I added new attributes, but not all of them, unlike the stage layer, it is no longer required to process every attribute, so for example attribute `locationSpecifics`, which is anonymized, and no stakeholder requested it, can be left out from this layer. The purpose of this object is to filter data on a granularity of attributes `messageType` and `eventId` and take the most recently processed record. This is done by ranking function combining `ROW_NUMBER` and `PARTITION BY`.

Another crucial role of the object is to provide every record with attributes `WS_flag` and `RetailFlag` which provide distinction between orders coming from retail and wholesale orders. To achieve it we need to take information from other core layer objects. That is a practice which is not used in most other layers, but there is an exception for the core layer since the alternative would mean a lot of duplicate processing.

New Object

While the previously mentioned `CL_TMagic_BuildUpAgreementEvent_MT` contains records regarding property owner decisions there are also records about different construction and installation agreements, which are just remotely related, have different structures, and cause the object to contain unnecessarily many NULL values.

To solve this issue, I created a new object in the core layer, called `CL_TMagic_PropertyOwnerDecision_MT`, its purpose is to contain strictly information about the landlord's decision, differentiating between NE3 and NE4. Instead of loading information from the stage layer, I utilized the processing done in the `buildUpAgreementEvent` core layer object, since the standardization and addition of `WS_flag` and `RetailFlag` would have to be repeated.

Unexpected complication

Right before I started working on the core layer, our team received a couple of emails from the business sides showcasing concerns about a few of the property owner decisions displaying different information compared to reality. We did an investigation of specific addresses and found out that the existing processing is correct, so the problem lies in not receiving some relatively small amount of data from the sourcing system.

This situation can happen and can be sometimes solved by ordering full loads of all the data available for the specific message type at regular intervals. But before implementing this method which would cause additional processing of significant amounts of data, I searched for the missing information in other message types, which we already receive. A bit surprisingly I located two completely different message types with information about property owner decisions. These decisions were not differentiating between NE3 and NE4 and they missed most of the other supplementary attributes, but the most crucial information which is the property owner's decision, and the address was there. Therefore, it was decided I could use these sources to substitute the missing information.

Solution

Combining different sources, which provide the same type of information is one of the responsibilities of a core layer. So, the new planned object was picked as a place of fusion for the 2 new sources with the already established one.

The first task was to stage the information from new sources, as it was located inside JSONs and not extracted yet, since it wasn't used up until now. Other than the decision status I staged addresses in the form of attributes `klsid_ps` and `folid_ps` and the date of the change of the status in the form of `status_modificationDate`, which was available just for one of the two new sources. The first one comes from message type "DigiOSS – FiberOnLocation" which is staged in `CL_TMagic_Stage_FiberOnLocation_MT` and the second one from message type "IBT – BuildingPart" stored in `CL_TMagic_BuildingPartResponse_MT`.

The new object implements full processing so the first step each day is to drop it and create a new one from scratch as visible in the figure below. The logic of the object is to prefer information coming from the original `BuildingPartEvent` and only in cases where there are more recent records with different information from other sources, should they be stored as well. For this purpose, I created a temporary table `latest_bpr_event`, with the most recently processed decision status for every address and a few other attributes for comparison to other sources.

```

DROP TABLE IF EXISTS ${Sdatabase.core_layer}$.CL_TMagic_PropertyOwnerDecision_MT;

CREATE ${Sexternal_tbl}$.TABLE ${Sdatabase.core_layer}$.CL_TMagic_PropertyOwnerDecision_MT
STORED AS PARQUET
${Slocation}$. "${Slocation_path}$.CL_TMagic_PropertyOwnerDecision_MT"
${Sstblproperties}$.
AS
WITH latest_bpr_event AS (
  SELECT folid_ps,
         klsid_ps,
         acl_dop_iso,
         eventtime_iso,
         agreementstatusne3,
         agreementstatusne4
  FROM ${Sdatabase.core_layer}$.CL_TMagic_BuildingPartEvent_MT
  WHERE acl_id_int IN (SELECT MAX(acl_id_int)
                       FROM ${Sdatabase.core_layer}$.CL_TMagic_BuildingPartEvent_MT
                       WHERE agreementstatusne4 IS NOT NULL
                       GROUP BY folid_ps, klsid_ps)
),

```

Figure 13 Full processing code and creation of temporary table for comparison (source: author)

The filtration of data from new sources is done in temporary WITH tables, starting with the fiberOnLocation source. I grouped them by address and decision status and added some complementary attributes such as the first and last date on which the specific status has appeared/changed, also I added the id and date of the latest processing. Additionally, I ranked the grouped records and displayed only the latest status for each address.

Then I joined the table with the latest_bpr_event on address attributes. Now I could compare the fiberOnLocation (fbl) source to the buildUpAgreementEvent source. In the WHERE clause, I set conditions for records that will be taken from fbl. First, the buildAgreeState, which contains the owner decision can't be NULL, because it wouldn't provide us with any value. Then latest modification date of fbl, which is an equivalent of event time, has to be higher than buildUpAgreementEvent's event time while also the property owner's decision has to be different in both sources.

The last two conditions were decided together with the product owner after studying the evolution of some concrete scenarios. The first of them is to take the owner's decision even if it is older than the most recent info, but only if the most recent info arrived on the same day and has the status "WAITING". The second rule is not to take fbl records with the status "WAITING" if the records from other sources display any of the "final" statuses ("APPROVED", "REJECTED") or they display status "POSTPONED" which is a value fbl never reaches and instead it just displays "WAITING".

```

fiberOnLocation AS
(
  SELECT giga.*
  FROM(
    SELECT folid_ps,
           klsid_ps,
           buildAgreeState,
           MAX(fol_ModificationDate_ISO) AS latest_statusType_date,
           MAX(acl_id_int) AS latest_acl_id_int,
           -- there are about 200 instances with duplicate acl_id_int, that's why we use acl_dop_iso as well
           MAX(acl_dop_iso) AS latest_acl_dop_iso,
           ROW_NUMBER() OVER (PARTITION BY folid_ps, klsid_ps ORDER BY MAX(acl_dop_iso) DESC) AS rnk
    FROM ${Sdatabse.stage_layer}$.CL_TMagic_Stage_FiberOnLocation_Response_MT
    GROUP BY folid_ps, klsid_ps, buildAgreeState
  ) giga
  LEFT OUTER JOIN latest_bpr_event latest
  ON latest.folid_ps = giga.folid_ps
  AND latest.klsid_ps = giga.klsid_ps
  WHERE buildAgreeState IS NOT NULL
  AND ((latest_statusType_date > eventtime_iso AND buildAgreeState != agreementstatusne4)
       OR latest.klsid_ps IS NULL
       -- there are occasions when we get approval and waiting status on the same day,
       -- then we prioritize approval even if it came earlier
       OR (CAST(latest_statusType_date AS date) = CAST(eventtime_iso AS date) AND
          buildAgreeState != agreementstatusne4 AND agreementstatusne4 = 'WAITING'))
  -- we prioritize the "final" status of approved or rejected over waiting
  -- and other sources seem to use waiting in place of postponed, since they dont have this status
  AND (agreementstatusne4 NOT IN ('APPROVED', 'POSTPONED', 'REJECTED') OR agreementstatusne4 IS NULL
       OR buildAgreeState != 'WAITING')
  AND rnk = 1
),

```

Figure 14 Creation of fiberOnLocation table for filtering data from new source and aggregation of date attributes (source: author)

The process is similar to buildingPartResponse (bpr). One of the differences is that we need to do standardization of values inside buildUpAgreementStatus because it shows “AGREED” instead of “APPROVED” and “PENDING” instead of “WAITING”. A further difference is an absence of event time or equivalent attribute, therefore I needed to use the second-best option, which is to compare the event time to the first time the information about specific status was processed. The last difference was that I no longer compared the data just to buildUpAgreementEvent but also to the already filtered fiberOnLocation table, to prevent showing the same status twice.

```

buildingPartResponse AS
(
...
SELECT bpr.*
FROM(
SELECT folid_ps,
klsid_ps,
CASE buildupagreementstatus
WHEN 'AGREED' THEN 'APPROVED'
WHEN 'REJECTED' THEN 'REJECTED'
WHEN 'PENDING' THEN 'WAITING'
ELSE NULL
END AS buildAgreeState,
MAX(acl_id_int) AS latest_acl_id_int,
MAX(acl_dop_iso) AS latest_acl_dop_iso,
MIN(acl_dop_iso) AS first_acl_dop_iso,
ROW_NUMBER() OVER (PARTITION BY folid_ps,klsid_ps ORDER BY MAX(acl_dop_iso) DESC) AS rnk
FROM ${Sdatabase.core_layer}$.CL_TMagic_BuildingPartResponse_MT
GROUP BY folid_ps, klsid_ps, buildupagreementstatus
) bpr
LEFT OUTER JOIN latest_bpr_event latest
ON latest.folid_ps = bpr.folid_ps
AND latest.klsid_ps = bpr.klsid_ps
LEFT OUTER JOIN fiberOnLocation fbl
ON fbl.folid_ps = bpr.folid_ps
AND fbl.klsid_ps = bpr.klsid_ps
WHERE bpr.buildAgreeState IS NOT NULL
AND (((first_acl_dop_iso > eventtime_iso AND bpr.buildAgreeState != agreementstatusne4)
OR latest.klsid_ps IS NULL)
AND ((first_acl_dop_iso > fbl.latest_statusType_date AND bpr.buildAgreeState != fbl.buildAgreeState)
OR fbl.klsid_ps IS NULL))
AND (fbl.buildAgreeState NOT IN ('APPROVED','POSTPONED','REJECTED') OR fbl.buildAgreeState IS NULL
OR bpr.buildAgreeState != 'WAITING')
AND (agreementstatusne4 NOT IN ('APPROVED','POSTPONED','REJECTED') OR agreementstatusne4 IS NULL
OR bpr.buildAgreeState != 'WAITING')
AND bpr.rnk = 1
),

```

Figure 15 Creation of buildingPartResponse table for filtering data from new source and aggregation of date attributes (source: author)

After this preparation the last step was to build the final object with all the required attributes, combining the information from 3 different sources. I created 3 tables taking data mainly from the original objects and restricting it by performing an INNER JOIN with the filtered tables. I also added a made-up attribute status_source, which shows the origins of the information. Finally, the 3 tables were combined using the UNION ALL command.

Data model



Figure 16 A simplified physical data model for CL_TMagic_propertyOwnerDecision_MT, created in Innovator (source: author)

3.4 Data Mart Layer

3.4.1 Broad Overview

The data mart layer represents a crucial component within an organization's data architecture, it serves as a focused repository of data tailored to specific business units or user groups. Unlike core layer objects, data marts are designed to provide targeted datasets that meet the unique analytical needs of departments such as sales, marketing, finance, or operations. This layer often acts as an intermediary between the core layer and end-users, offering pre-summarized, structured data optimized for quick querying and reporting. By organizing data at this granular level, data marts enable faster access to relevant information, empowering decision-makers to derive actionable insights quickly and efficiently. (Inmon, 2005)

3.4.2 Types of Data Marts

Dependent Data Mart

- A dependent data mart obtains its data directly from an enterprise data warehouse or a centralized data repository.
- This type of data mart relies on the previous layer for data integration and transformation processes.
- The advantage is that most of the data administration happens in the central repository, so users of the data mart require less technical knowledge to operate it.

Independent Data Mart

- An independent data mart is built separately from an enterprise data warehouse and can source data directly from operational systems or other data sources.
- This type of data mart has its own data integration and transformation processes, which can be less complex and more focused compared to a dependent data mart.
- Their main use case is for business units that require more control over their data and reporting requirements. (Morris, 2021)

Hybrid Data Mart

- Hybrid data marts combine characteristics of both dependent and independent types, making them suitable for organizations with diverse databases requiring rapid responsiveness.
- A hybrid data mart allows users to integrate data from operational source systems and a data warehouse, ideal for ad hoc integration needs like incorporating new business groups or products.
- It can be more efficient to use this model as a transition step for new data marts, since new subject-specific data sources may be more easily managed as partially independent data marts.
- A complication might be the need for technical expertise at both the central data warehouse and data mart layer. (Naeem, 2019)

3.4.3 Implementing Business Logic

The data mart layer serves as a critical location for the implementation and integration of business logic into the data environment. This layer is where business rules, calculations, and transformations are applied to preprocessed core layer data to derive meaningful insights and metrics tailored to specific organizational needs. Business logic added within the data mart layer ensures that data is processed and presented in a way that aligns with business objectives and requirements. This may involve implementing custom calculations, defining data validations, comparing data values and dates, or creating derived attributes based on complex business rules. While it is possible to stage data and do all the processing in the core layer just based on principles of having all data cleaned, standardized, and stored in some logical objects, to create a data mart data engineer needs to know and understand concrete business requirements and business rules. (Inmon, 2005) By incorporating business logic at the data mart layer, organizations can centralize and standardize their analytical processes, enabling consistent and accurate decision-making across different business functions and user groups. Moreover, this approach enhances the agility and responsiveness of data analytics, allowing for iterative refinement of business logic to answer evolving business challenges and opportunities. (Hughes, 2013)

3.4.4 Access and Governance

Access to the data mart layer varies based on user roles and organizational requirements.

Data experts, such as data engineers and analysts, often have comprehensive access for querying and maintenance, while business users typically interact with predefined reports and tables. Access controls include role-based access, authentication, and governance policies to ensure security and compliance. (Ladley, 2012) In some cases, the data mart layer serves as the final layer for end-user access, while in other architectures, there may be additional layers like an export layer for data distribution to external systems or partners. In the first scenario it makes sense that business users can at least view data on the data mart layer, whereas if the export layer exists, data marts are usually protected from any outside access other than the technical experts. This layered approach enables organizations to manage access effectively. (Plotkin, 2021)

3.4.5 Implementation

In our project, we use dependent data marts which rely on already preprocessed data from the core layer. When it comes to access to the data, we don't let our stakeholders attain it directly from data marts as we have an export layer which serves this purpose.

The main objective of the implementation in this layer is to incorporate newly acquired data about property owner decisions into one of the most crucial data marts called CL_TMagic_DM_distributionPoint_MT. It is on a granularity of one building and its primary identifier is folid_ps, the pseudonymized address of the building.

The first step was to summarize the data coming from the core layer object where we get multiple records per one address. This was done in a separate WITH table via the GROUP BY command on the fol id level. In this table, I also created new date attributes for the latest property owner decision and reaction. The difference between the two is that the decision doesn't include values postponed and waiting. Last but not least I determined what is the most recent property owner reaction for NE3 and NE4.

With this preparation, I can move on to the next table, where additional business logic is implemented. There are 4 more sources which we have to consider when determining the state of the landlord's approval. First is a table cl_FIA_REF_ftth_htn_mt which we receive from our colleagues in the fiber analytics team. The second is from the table regarding orders, the third is from the table about the connection of the building, and the fourth is about special types of orders.

```

FROM all_distrP_with_area_info distrP
LEFT OUTER JOIN latestPropertyOwnerDecision pod ON pod.folid_ps = distrP.folid_ps
LEFT OUTER JOIN orderData o ON o.folid_ps = distrP.folid_ps
LEFT OUTER JOIN connection_only co ON co.folid_ps = distrP.folid_ps
LEFT OUTER JOIN CO_netnet ON CO_netnet.folid_ps = distrP.folid_ps
LEFT OUTER JOIN wowi_apartment_order wowi ON wowi.folid_ps = distrP.folid_ps
LEFT OUTER JOIN (SELECT kls_id_ps, approved_at, wowi_produkt, source
FROM (
SELECT kls_id_ps,
approved_at,
wowi_produkt,
source,
ROW_NUMBER () OVER (PARTITION BY kls_id_ps ORDER BY approved_at DESC) AS rnk
FROM ${Sdatabase.core_layer}$.cl_FIA_REF_ftth_htn_mt
WHERE source in ('WOWI','IV','IV AND WOWI','POM')
)a
WHERE a.rnk = 1
) FIA
ON FIA.kls_id_ps = distrP.klsid_ps

```

Figure 17 Joining additional sources for landlord's decisions in DM (source: author)

Below we can see all the different conditions that can lead to a change in the property owner's decision status. Without getting into too much detail I'll explain the main ones. First, the sole occurrence of the address in the FIA table corresponds with the building having an NE3 agreement, to determine if NE4 is approved as well, we must look at attribute wowi_produkt. Next up are several orders from table connection only, if there are any, we can assume that the owner gave NE3 approval, but we can't make any conclusions about NE4.

Subsequently, we also analyze data about tariff orders, if the latest order was created more recently than the latest information from our main property owner decision source, then we consider the owner decision as approval for both NE3 and NE4. Last but not least we take into account wowi orders for apartments, if there are any it is automatically accounted for as owner's approval for NE3 and NE4.

```

CASE
  ....-- FIA HTN info
  .... WHEN FIA.kls_id_ps is not null
  .... THEN 'confirmed'
  ....-- WOWI confirmed
  .... WHEN NVL(co.num_order_c_net_WOWI, 0) + NVL(wowi.number_of_order_wowi_apartment_net, 0) > 0
  .... THEN 'confirmed'
  ....-- Order owner gross (AVIA cancelation != property owner rejection)
  .... WHEN NVL(o.date_latest_create_order_t_gross_owner, '1900-01-01 00:00:00') > pod.Date_latest_Property_Owner_Reaction_NE3
  .... OR NVL(co.date_latest_create_order_c_gross_owner, '1900-01-01 00:00:00') > pod.Date_latest_Property_Owner_Reaction_NE3
  .... THEN 'confirmed'
  ....-- latest reaction = confirmed
  .... WHEN pod.latest_NE3_Reaction = 'APPROVED'
  .... THEN 'confirmed'
  ....-- if ne4 approved ne3 as well (few cases where ne3 is postponed while ne4 approved, which is not possible)
  .... WHEN pod.latest_NE4_Reaction = 'APPROVED'
  .... THEN 'confirmed'
  ....-- latest reaction = rejected
  .... WHEN pod.latest_NE3_Reaction = 'REJECTED'
  .... THEN 'rejected'
  ....-- pending
  .... WHEN NVL(o.number_of_order_net, 0) + NVL(co.netnet.number_of_co_net_net, 0) > 0
  .... AND (pod.latest_NE3_Reaction IS NULL OR pod.latest_NE3_Reaction = 'WAITING')
  .... THEN 'pending'
  ....-- postponed
  .... WHEN pod.latest_NE3_Reaction = 'POSTPONED'
  .... THEN 'postponed'
  ....-- no order tc net and no decision
  .... WHEN NVL(o.number_of_order_net, 0) + NVL(co.netnet.number_of_co_net_net, 0) = 0
  .... AND (pod.latest_NE3_Reaction IS NULL OR pod.latest_NE3_Reaction = 'WAITING')
  .... THEN 'no agreement, no net presales order'
  .... ELSE 'undefined'
END
.....AS Property_Owner_Decision_NE3,

```

Figure 18 Case statement for creation of property owner NE3 decision (source: author)

Similar logic to the one which was used for attributes Property_Owner_Decision_NE3 and Property_Owner_Decision_NE4 is also used for their date attributes, as all the utilized sources provide dates of the owner's reaction, the only challenge was to determine which one should have a preference in scenarios where a combination of more sources was used.

```

IF(approved_at is not null, approved_at,
  .... IF(NVL(co.num_order_c_net_WOWI, 0) + NVL(wowi.number_of_order_wowi_apartment_net, 0) > 0,
  .... LEAST(COALESCE(co.date_first_create_order_c_net_WOWI, wowi.date_first_create_order_w_net),
  .... COALESCE(wowi.date_first_create_order_w_net, co.date_first_create_order_c_net_WOWI)),
  .... IF (o.date_latest_create_order_t_gross_owner > NVL(pod.Date_latest_Property_Owner_Reaction_NE3, '1900-01-01 00:00:00')
  .... OR co.date_latest_create_order_c_gross_owner > NVL(pod.Date_latest_Property_Owner_Reaction_NE3, '1900-01-01 00:00:00'),
  .... GREATEST(COALESCE(o.date_latest_create_order_t_gross_owner, co.date_latest_create_order_c_gross_owner),
  .... COALESCE(co.date_latest_create_order_c_gross_owner, o.date_latest_create_order_t_gross_owner)),
  .... pod.Date_latest_Property_Owner_Decision_NE3)
  .... )
)
.....AS Date_Property_Owner_Decision_NE3,

```

Figure 19 Selection of correct date of owner NE3 decision (source: author)

The owner's decisions and their dates were the main attributes regarding our topic, and they were stored together with a couple of complementary attributes in the large data mart distribution point, from where they can be accessed by other data marts and exports.

3.5 Export Layer

3.5.1 Broad Overview

The export layer in a data structure refers to a stage where processed and refined data is gathered, before being exported to consumers or downstream systems. This layer acts as a final staging area where data is prepared for consumption, ensuring that it meets specific requirements or standards set by the receiving applications or users. The export layer may involve formatting data into different schemas, applying data transformations or aggregations, and enriching data with relevant metadata or contextual information. In an ideal state, the export layer should be very simple. The main focus is on collecting data from one or more data marts and if necessary, performing some final touches, but most of the processing should be already done in previous layers. The export layer plays a pivotal role in ensuring that the exported data is accurate, meaningful, and optimized for its intended use cases or applications. (Kimball & Ross, 2013)

3.5.2 Implementation

In our scenario, we had about 15 different exports that included the general property owner decision and since the migration to the divided decision caused the removal of the general one, I had to determine what type of decision should be appointed in which object. This created a slight complication, because communicating with all of the affected stakeholders and waiting for them to decide if they prefer NE3, NE4, or both could potentially last many weeks.

To find an alternative I discussed the matter with our product owner who was able to make decisions for the majority of affected objects (there are not just exports, but data marts affected as well). Afterwards, I was left with just a handful of exports which needed further communication with stakeholders to find their desired replacement.

In the end, I had a list of all the objects which would use NE4 approval and just a connection only data mart and its exports preferred NE3 decision. Also, we concluded that we won't provide the possibility of having both NE3 and NE4 decisions in exports for now and later on in the summer when the migration to the Google Cloud platform will be completed and the team has more resources, we can offer the additional attribute as well. This simplifies the deployment process significantly as we don't have to adjust any of the many agreements we have with business sides regarding those exports.

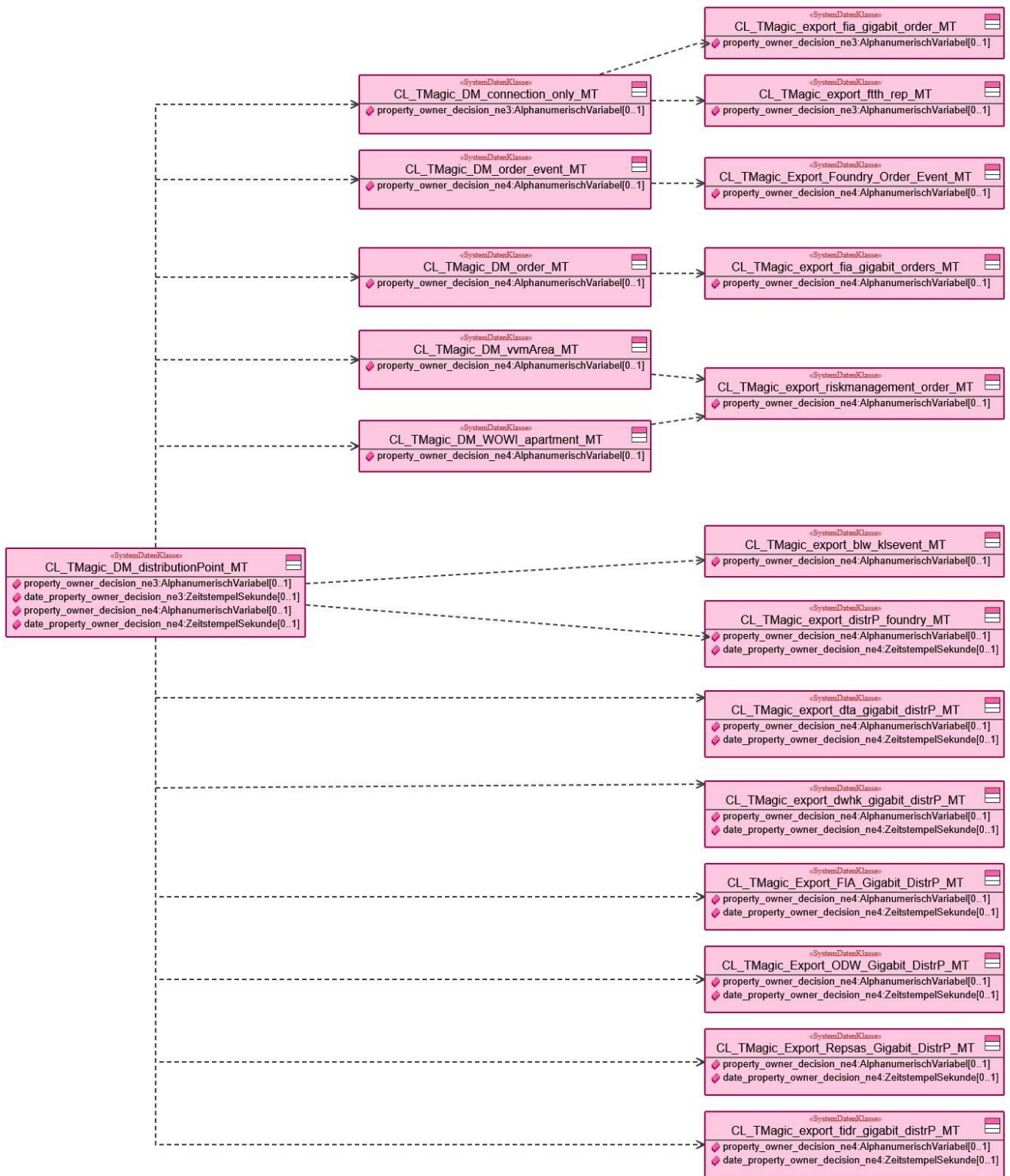


Figure 20 Data model for illustration of all the exports using one of the property owner's decisions (source: author)

3.6 Chapter Conclusion

This chapter took readers through the whole data journey, starting with a collection of raw data, followed by the staging layer, the core layer, and the data mart layer all the way up to the export layer. Each layer was firstly characterized more on the theoretical level, which gave readers a better understanding of what were the concrete steps the author took in the implementation parts of the thesis.

At the beginning of the chapter, the author had just a description of the business requirements provided by the product owner. Throughout this chapter author documented the whole process of acquiring and processing data to achieve a state where the company has continuously updated data about NE3 and NE4 decisions in exports and stored in the data mart for any future usage. What's missing is fulfilling the business needs for new concrete KPIs about property owner decisions and ways to visualize these values. This will be the subject of the next chapter.

4 Visualization

Data visualization is a powerful tool for transforming complex datasets into intuitive and actionable insights. This chapter will explore the principles and techniques of data visualization, highlighting its significance in modern data-driven organizations. Data visualization enables stakeholders to understand trends, patterns, and relationships within data more effectively than raw numbers or text alone. By leveraging visual elements such as charts, graphs, and dashboards, organizations can communicate information clearly and compellingly, facilitating data-driven decision-making at all levels. From discussing different visualization types and design principles to showcasing tools and technologies for creating impactful visuals, this chapter provides a comprehensive overview of data visualization's role in turning data into actionable intelligence. Similarly to the last chapter, it begins with laying some theoretical groundwork and then the author will go through concrete steps to achieve visualization of the landlord's decisions in the form of a dashboard.

4.1 Types of Visualizations

There are plenty of various types of data visualizations. They all serve different purposes and should be used according to the nature of the data and the message author wants to convey. It can be viewed just like a set of craftsman's tools, sometimes there is more than one tool that could get the job done, but there is usually one which is the most effective. And a skilled craftsman knows which tool is right for each scenario.

Here is a list of a few of the most common visualization types and their intended usage.

- Bar Chart:
 - Uses rectangular bars to represent categories or values.
 - Useful for comparing quantities across different categories.
- Line Chart:
 - Displays data as a series of data points connected by straight lines.
 - Ideal for showing trends or changes over time.
- Pie Chart:

- Divides a circle into segments to represent the proportions of a whole.
- Suitable for showing the distribution or composition of a dataset.
- Scatter Plot:
 - Uses dots to represent individual data points based on two variables.
 - Helpful for visualizing relationships or correlations between variables.
- Histogram:
 - Displays the distribution of numerical data through bars.
 - Shows the frequency or count of data within predefined intervals (bins).
- Heatmap:
 - Represents data values in a matrix using colors to visualize variations.
 - Often used to show patterns or correlations in large datasets.
- Area Chart:
 - Similar to a line chart but uses filled areas below the line to represent data.
 - Suitable for visualizing cumulative changes over time. (Velarde, 2024)
- Standard text formatting:
 - Displays just a single number with plain text explaining the value.
 - Effective for showing values that don't need additional context. (Bock, 2017)

4.2 Principles of Visualization

Data visualization principles encompass a set of guidelines and best practices for designing effective and impactful visual representations of data. These principles emphasize clarity, simplicity, and accuracy in conveying complex information visually. (Kumar, 2024) Key principles include choosing appropriate chart types based on the data and the intended message, using meaningful and consistent labeling and color schemes, and ensuring that visualizations are easy to interpret and understand at a glance. Effective data visualization

should prioritize the audience's needs and goals, and always take into account the level of expertise the audience has on the topic. Other principles focus on avoiding clutter and unnecessary information, focusing on the story the data tells, and maintaining integrity and transparency in data representation. By adhering to these principles, data visualizations can become powerful tools for communication, analysis, and decision-making across various domains and industries. (Cairo, 2016)

4.3 Implementation

To prevent having to perform complex calculations, grouping functions, and joining tables in the visualization application, we delegate these responsibilities to a separate object where the KPIs are calculated, and which can be used as a single source of information for an entire dashboard.

4.3.1 KPIs calculations

I created the object `CL_TMagic_KPI_propertyOwnerDecision_MT`, its main source is the data mart distribution point that is on the granularity of a single building. I grouped the data by several attributes, for example, the vvm areas, these groups can be later on incorporated as filtering options.

There were clear requirements on the precise KPIs our stakeholders requested. They could be separated into two groups first one was focused on numbers and percentages of buildings with different owner reactions. The second one a bit bigger and more complex focuses on HHUstos which is a combination of business units and living units inside of one building.

The business side is interested in 4 scenarios for both HHUstos and buildings, first is when NE3 is rejected, second is no reaction for both NE3 and NE4, third is approval for NE3 but rejection or open status for NE4, and lastly approval for NE4 (which inherently means NE3 approval as well).

In addition, there should be two KPIs distinguishing between HHUstos with no reaction which don't have a tariff order, and those who do but are also not connected yet. These same two KPIs also need to be created for NE4 approval. The last KPI will show the number and percentage of HHUstos with NE4 approval that are already connected to fiber.

4.3.2 Dashboard creation

The application we use for creating visualizations is MicroStrategy. As for the form of the visualization, MicroStrategy offers dossiers which are newer versions of dashboards and are supposed to differ by being more interactive and collaborative.

To follow the principles of proper visualization I looked at who's intended audience for this dashboard. The answer is one knowledgeable stakeholder who requested 15 precise numbers and 13 related percentages. He understands what these KPIs represent and therefore the dashboard doesn't need a lot of explanations. Also, he didn't ask for any charts or comparisons and therefore it's clear that the solution is just plain KPIs.

Another principle I adhered to was trying to keep the dashboard as simple as possible, with minimal text, and conveying the message also via the choice of the right colors and positioning of the KPIs.

When it comes to filtering our stakeholder wanted the ability to focus on specific areas, regions, expansion types, hierarchy groups, and years of the rollout program. I provided these filtering options in the right sidebar, where each filter has its check box with the ability for multiple selections and a search bar for finding specific results.

The accuracy of the information inside this dashboard can be easily checked by doing a couple of test equations, for example, the summarization of all the buildings with different statuses has to equal the number of all the buildings in our system, and the same holds true for HHUstos. HHUstos with no reaction and with NE4 approval have their own subcategories and they as well when summed up have to be equal to their superior value. A bit faster alternative is to check the percentages which when summed up according to their groups should always result in 100 %.

For security reasons, I cannot display the actual data in the dashboard below and therefore I filled it with artificially created dummy data.



Number of buildings in GGS		Number of HHUstos in GGS			
4,972,618		14,481,731			
Buildings with NE3 rejection		HHUstos with NE3 rejection			
43,079		192,877			
0.9%		1.3%			
Buildings with no reaction		HHUstos with no reaction			
2,262,557		6,893,353			
45.5%		HHUstos without order with no reaction		Net-Tariff-orders with no reaction	
		119,689		6,773,664	
		1.7%		98.3%	
Buildings with NE3 approval only		HHUstos with NE3 approval only			
35,927		304,346			
0.7%		2.1%			
Buildings with NE4 approval		HHUstos with NE4 approval		Connected HHUstos	
2,631,055		7,091,155		49.0%	
		HHUstos without order with NE4 approval		Net-Tariff-Orders with NE4 approval	
		5,656,570		1,434,585	
		79.8%		20.2%	
52.9%				20.3%	
				1,436,999	

Figure 22 Property owner decision dashboard without filter (source: author)

4.3.3 Results

Now as I have gathered all the data is the right time to come back to the original question of the thesis, which is how many approvals does the telecom company have for fiber connection just to the basement and how many to the living units. Or in other words how many buildings have NE3 status as confirmed and how many have NE4 approval.

I'll provide two answers one for all of the gathered records we've been receiving for over 3 years and one for the time period between 25.11.2023 up until 21.4.2024. The reasoning being that I subscribed to version 2 of buildUpAgreement messages in the fall of 2023, so our main source of data regarding the distinction between NE3 and NE4 is active only for the above-mentioned time interval.

Results for all records that we have available:

- NE3 approvals – 2 303 914
- NE4 approvals – 2 260 701
- difference – 43 213

Results for records available since 25.11.2024:

- NE3 approvals – 477 081
- NE4 approvals – 465 003
- difference – 12 078

It is apparent that in the majority of cases, landlords agree to the rollout of fiber to both the basement and to the living unit. Only in about 2,5 % of cases did landlords decide to connect just the Onebox (device for further distribution of fiber into the building) in the basement. The lower this percentage is the better for the company in terms of potential new customers and also lower costs, since change of mind of the owner and delayed approval leads to increased costs for returning to the already connected area.

Conclusion

The main goal of this thesis has been successfully reached as the question of approval rates for the basement and the living unit has been answered. The results display that the overwhelming majority of landlords who agree to fiber roll-out tend to let the telecom company configure the fiber connection not only to the basement but also to the living unit. This thesis did not judge how satisfactory these results are. Further research could be done in comparison to other telecom companies and what are some factors and mechanisms to increase the approval rate.

Moreover, this study has delivered an examination of the telecommunication landscape, coming to the conclusion of fiber optic superiority in most of its attributes compared to its predecessor. It followed up with an analysis of the current status regarding fiber roll-out in Germany, which turned out to be behind other countries and in the middle of intensive and costly efforts to reach full fiber connectivity by 2030.

On the way to reach the main objective this thesis also delivered the partial goal of providing a detailed exploration of the data journey spanning various layers of a structured data architecture, from data acquisition through the staging layer, core layer, and the export layer, the research has shed light on the complexities and transformations inherent in telecom data management.

Furthermore, the development and deployment of a daily updated dashboard have facilitated the effective visualization and communication of findings. This combined with the unexpected fix of the incomplete data, incorporation of newly divided landlord's decisions into the existing architecture, and overview of the whole data process, brings substantial value to the telecom company.

In conclusion, this thesis aims to deliver value to its readers by offering an interesting introduction to fiber optics technology and its crucial role in the telecommunications industry. Additionally, the documentation of the data journey through various architectural layers provides insights into modern data management practices. Moreover, this work serves as a practical example of real-world big data tasks, demonstrating the complexities and methodologies involved in processing and analyzing telecommunications data. It is the author's aspiration that this thesis not only contributes to academic knowledge but also serves as a resource for starting professionals interested in telecommunications, data engineering, and the evolving landscape of fiber optic technology.

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